Quiz 3: Fields & Waves (ECE230), Winter 2024

Duration: 1hr, Total: 15 points (Attempt all questions)

**Vector quantities must be represented with an overhead arrow, vector calculus operators must be written correctly, dot and cross products must be written properly. In case of any of these mistake -1 will be awarded in each question with such mistake.

NO credit will be given to the answers that do not accompany proper explanation.

Any case of copying/cheating will be dealt as per institute guidelines.

Q1. Consider a wire of radius r carries a current I_0.

(a) If the current is uniformly distributed over the surface what is the surface current density?

(b) If the current is distributed in such a way that the volume current density is inversely proportional to the distance from the axis of the wire, what is the volume current density?

[$2+3=5$ points]

[Note that wrong units will lead to marks deduction]

Q2. Consider a $10cm \times 5cm$ rectangular loop (single turn) with a resistance of 0.4Ω is placed in the $xy-$plane. The region has a spatially uniform, time varying magnetic field $\frac{1}{4}\sin(100\pi t)\hat{z}$. What is the power dissipated in the loop?

[4 points]

Q3. Consider a plane wave propagation in free space with an electric field: $\vec{E} = \hat{z}E_0 \exp\left[i\left(1.5\pi \times 10^9 t - 3\pi x - my\right)\right]$.

(a) Determine the wavelength.

(b) What angle does the direction of propagation make with the x-axis?

[$3+3=6$ points]